

Special Senses – Eye & Ear

Laboratory Experience

The purpose of this laboratory exercise is to identify the different tissues and cells found in the eye and ear. Utilize the slides listed below along with the lab supplement, text and atlas to identify the cells, tissues & structures of the sensory organs.

I. Eye

General Structure

Slide #2 Diagram – layers of the eye

Slide #3 Modified drawing of the eye

Slide #4 Layers of the eye, H&E

The wall of the eye consists of three layers or coats. The **corneoscleral coat** is the outer fibrous layer that includes the sclera and cornea. The **vascular coat** is the middle layer or uvea that includes the choroid, ciliary body and iris. The **retina** is the inner layer that includes the retinal pigment epithelium (RPE) and neural retina.

The layers of the eye and the lens serve as the boundaries for three chambers of the eye. The **anterior chamber** is the space between the cornea and iris. The **posterior chamber** is the space between the iris and lens. The **vitreous chamber** is the space between the lens and neural retina.

The eye can also be divided into an anterior segment and posterior segment. The **anterior segment** includes the cornea, anterior & posterior chambers and their contents. The **posterior segment** consists of the vitreous chamber, visual retina, RPE, posterior sclera and uvea.

Retina

Slide #5 Diagram – layers of the retina and H&E x440

Slide #6 Retina, H&E x325

Slide #7 Optic disc, H&E x65

The innermost layer of the eye is the **retina** and consists of two layers: **neural retina and RPE**. There is a potential space between the two layers. Separation of the layers or retinal detachment can occur because of trauma or other pathological conditions. The RPE is attached to the Bruch's membrane of the choroid. The neural retina has two regions. The **nonphotosensitive region** is located anterior to the **ora serrata** and lines the ciliary body and iris. The **photosensitive region** (pars optica retinae) lines the inner surface of the eye posterior to the ora serrata. The pars optica retinae (normally just called the retina) is organized into ten layers.

1. **Retinal pigment epithelium (RPE)** - simple cuboidal epithelium attached to Bruch's membrane. Melanin pigment is present in the apical cytoplasm. Microvilli extend between the processes of the rods and cones. RPE phagocytose the shed discs from the rods and cones. RPE have tight junctions serve as major component in the blood-retina barrier.
2. **Layer of rods and cones** – contains the outer and inner segments of the photoreceptor cells. Outer segments contain the photosensitive membranous discs.
3. **Outer limiting membrane** – apical boundary of Müller's cells. A row of zonulae adherents that attaches Müller's cells to each other and to the rods and cones
4. **Outer nuclear layer** - cell bodies (nuclei) of the rods and cones.
5. **Outer plexiform layer** – processes of the rods and cones; and the processes of horizontal, amacrine and bipolar cells that connect to them.
6. **Inner nuclear layer** – contains the cell bodies (nuclei) of horizontal, amacrine, bipolar and Müller's cells.
7. **Inner plexiform layer** – contains the processes of horizontal, amacrine, bipolar and ganglion cells that connect to each other.
8. **Ganglion cell layer** - contains the cell bodies (nuclei) of ganglion cells. Ganglion cells are large multipolar neurons. Dendrites extend to inner plexiform layer. Axons pass into the layer of optic nerve fibers. Small neuroglia cells are scattered among the ganglion cells.
9. **Layer of optic nerve fibers** – contains the axons of ganglion cells that lead from the retina to the brain. The axons are thin, nonmyelinated processes. This layer increases in depth as the axons converge at the optic disc. Retinal vessels are primarily located in this layer.
10. **Inner limiting membrane** – composed of the basal lamina of Müller's cells. It separates the retina from the vitreous body.

Corneoscleral Coat

Slide #8 Anterior eye, H&E

Slide #9 Cornea, H&E x175; Corneal stroma, TEM x16700

Slide #10 Corneal epithelium and endothelium, H&E x360

Slide #11 Anterior eye, H&E x45

Slide #12 Corneoscleral junction, H&E x130; Corneoscleral junction and canal of Schlemm, H&E x360

The corneoscleral coat consists of the cornea and sclera. The **cornea** consists of five layers.

1. **Corneal epithelium** - stratified squamous nonkeratinized. About 5 layers of cells and continuous with the conjunctival epithelium. The most apical layer of cells exhibits microvilli (vermiform ridges) that retain a film of tears over the corneal surface. Numerous free nerve endings provide an extreme sensitivity to touch. Corneal epithelium stem cells reside at the corneoscleral limbus. The corneal epithelium has good regenerative capability with a turnover time of about 7 days.
2. **Bowman's membrane** - anterior basement membrane for the corneal epithelium. It lies between the corneal epithelium and the corneal stroma; provides some strength and acts as a barrier. Bowman's membrane cannot regenerate; if damaged, a scar forms that may impair vision.
3. **Corneal stroma** (substantia propria) - the thickest part of the cornea composed of about 60 thin lamellae. Each lamellae consists of parallel bundles of collagen fibrils. Between lamellae are flattened corneal fibroblasts (corneal keratocytes). The collagen fibrils in each lamella are arranged at approximately right angles to those in the adjacent lamellae (orthogonal array). Ground substance contains corneal proteoglycans.
4. **Descemet's membrane** - posterior basement membrane for the corneal endothelium. It is unusually thick (up to 10 μm) and capable of regeneration after injury.
5. **Corneal endothelium** - simple squamous epithelium lining the inner surface of the cornea. It provides for metabolic exchange between the cornea and aqueous humor. Na^+/K^+ ATPase pumps located in the lateral plasma membrane function to regulate water content of the stroma to help maintain transparency. Corneal endothelium has a limited regenerative capability.

The corneoscleral limbus is the transition zone between the cornea and sclera. The corneal epithelium transitions to the conjunctival epithelium. The iridocorneal angle of the limbus region contains the apparatus for the outflow of aqueous humor. In the stroma, endothelial-lined channels merge to form the scleral venous sinus (canal of Schlemm).

The **sclera** is dense irregular connective tissue (type I collagen fibers) that strengthen and support the wall of the eye. It is continuous with the dura mater of the optic nerve.

Vascular Coat (Uvea)

Slide #13 Anterior eye, H&E x480

Slide #14 Iris, diagram and H&E x570

The vascular coat consists of the choroid, ciliary body & iris.

The **choroid** is a dark brown vascular sheet between the retina and sclera. It consists of the **choriocapillary layer** and **Bruch's membrane**. The choriocapillary layer is the inner layer of vessels arranged that provides nutrients to the cells of the retinal layers 1-5. Bruch's membrane (lamina vitrea) lies between the choriocapillary layer and the retinal pigment epithelium. It is a multilaminar sheet consisting of a center elastic layer and two layers of collagen fibers. At the ora serrata, it becomes continuous with the basal lamina of the ciliary body.

Ciliary body

The **ciliary body** is the thickened anterior portion of the vascular coat. The anterior third of the ciliary body has about 75 radial ridges or **ciliary processes**. The zonular fibers arise from the grooves between the ciliary processes. The stroma is organized into two layers. The outer layer consists of smooth muscle, the **ciliary muscle**. The inner vascular region extends into the ciliary processes. The ciliary body and ciliary processes are covered by a double layer of ciliary epithelium.

The ciliary muscle contains smooth muscle bundles that function to alter the shape of the lens for near or distant vision. The contraction of the circular bundles reduces the tension on the zonular fibers resulting in the lens becoming more spherical (**accommodation**). The relaxation of the bundles increases the tension on the zonular fibers resulting in the lens becoming flattened. The ciliary muscle is innervated by parasympathetic fibers.

Ciliary processes are ridge-like extensions of the ciliary body. The ciliary processes are covered by a double layer of ciliary epithelium – the inner nonpigmented layer and outer pigmented layer. The ciliary epithelium has three major functions: secretion of the aqueous humor; help maintain the blood-aqueous barrier; and secretion & anchoring of zonular fibers. Aqueous humor is formed from the plasma passing through fenestrated capillaries of the stroma and the active secretion by the nonpigmented epithelium. The major component of the zonular fibers is fibrillin. The zonular fibers form a ciliary zonule (of Zinn) or band that connects the ciliary body to the lens.

Iris

The **iris** is the most anterior portion of the vascular coat and forms a contractile diaphragm in front of the lens. The **pupil** is the central aperture of the iris. The size of the pupil is controlled by contraction of the sphincter and dilator pupillae muscles. The iris consists of a vascular stroma, a layer of epithelium and a layer of myoepithelial cells.

The stroma of the iris contains loose connective tissue with smooth muscle cells, fibroblasts and melanocytes. The number of pigmented cells affects an individual's eye color. Few pigmented cells result in a blue color; intermediate results in a green or amber color; high amount results in a brown or dark brown color.

Constriction of the pupil is produced by smooth muscle located in the stroma near the pupillary margin called the **sphincter pupillae muscle**. The sphincter pupillae muscle is innervated by parasympathetic fibers.

The posterior surface of the iris is covered by a layer of highly pigmented cells, the **posterior pigment epithelium**. The intense pigmentation prevents the visualization of most cellular structures. Located anterior to the posterior pigment epithelium is a layer of myoepithelial cells, the **anterior pigment myoepithelium**. The anterior portions of the myoepithelial cells have

processes that extend radially and collectively form the **dilator pupillae muscle**. The dilator pupillae muscle is innervated by sympathetic fibers.

Lens

Slide #15 Lens, diagram and H&E x570

Slide #16 Lens, H&E x360

The crystalline lens is a transparent, avascular, biconvex structure suspended between the edges of the ciliary body by the zonular fibers. The lens has three components: lens capsule, subcapsular epithelium and lens fibers. The **lens capsule** is a thick basal lamina produced by the anterior lens cells. The **subcapsular epithelium** is a cuboidal layer of lens epithelial cells present only on the anterior surface. The **lens fiber cells** are the long, thin cells without organelles derived from the lens epithelial cells. Epithelial cells proliferate at the germinal zone at the lens equator. They migrate and differentiate into mature lens fiber cells.

Eyelid

Slide #17 Eyelid, diagram and Picric acid x20; inset x60

The primary function of the eyelid is to protect the eye. The **palpebral conjunctiva** lines the undersurface of the eyelid and is continuous with the conjunctiva of the eye (bulbar conjunctiva). The palpebral conjunctiva consists of nonkeratinized stratified epithelium with goblet cells that can vary from stratified columnar to stratified squamous depending on the location.

Within each eyelid is a flexible support called the **tarsal plate**. A smooth muscle, **superior tarsal muscle**, attaches to the tarsal plate. Embedded within the tarsal plate are long sebaceous glands called **tarsal glands (Meibomian glands)**. The **orbicularis oculi muscle** overlies the tarsal plate.

Eyelashes emerge from the lid margin. **Sebaceous glands (of Zeis)** and **apocrine glands (of Moll)** are associated with the hair follicles.

Identify the major components of the eye and associated structures: Corneoscleral coat, sclera, cornea, vascular coat, choroid, ciliary body, iris, pupil, retina, neural retina, retinal pigment epithelium, lens, posterior chamber, anterior chamber, vitreous body, optic nerve, fovea centralis, ora serrata, eyelid, tarsal glands, eyelashes, sebaceous glands, apocrine glands, orbicularis oculi and conjunctiva.

II. Ear

General Structure

Slide #18 Diagram – divisions of the ear

Slide #19 Diagram – sensory regions of membranous labyrinth

Slide #20 Horizontal section of temporal bone, x65

The ear is a three-chambered sensory organ of hearing and equilibrium. The **external ear** collects, amplifies, and conducts sound to the tympanic membrane (eardrum). The **middle ear** is

an air-filled space that contains three small bones, the **auditory ossicles**. The ossicles connect the tympanic membrane to the oval window. The internal ear consists of three connected spaces of the bony labyrinth (**vestibule, cochlea and semicircular canals**) and their associated membranous labyrinth structures. The vestibule is the central space that contains the utricle and saccule. The cochlea extends anteriorly and contains the cochlear duct and organ of Corti. The semicircular canals extend posteriorly and contain the cristae ampullaris.

Utricle and semicircular canals

Slide #21 Diagram – crista ampullaris of semicircular canal

Slide #22 Diagram – macula of utricle

Slide #23 Horizontal section of temporal bone, H&E x20; Crista ampullaris, H&E x380; Macula of utricle, H&E x380

Slide #24 Internal ear, H&E x20

Slide #25 Semicircular canal, H&E x85, inset x380

The membranous labyrinth of the vestibule contains the macula of the utricle and macula of the saccule – structures that sense the position of the head and linear movement (static equilibrium).

The membranous labyrinth of the semicircular canals contains three cristae ampullaris - structures that sense the angular acceleration (turning) of the head (dynamic equilibrium).

Cochlea

Slide #26 Cochlea, diagram and H&E x65

Slide #27 Cochlea, H&E x65, inset x380

Slide #28 Organ of Corti, diagrams and SEMs x3250, x2400

Slide #29 Organ of Corti, diagram and H&E x180

The cochlea contains the cochlear duct (scala media) and the spiral organ of Corti – receptor for sound transduction (hearing).

Identify the major components of the ear and associated structures: external ear, tympanic membrane, middle ear, malleus, incus, stapes, internal ear, vestibule, macula of utricle, semicircular canals, cristae ampullaris, cochlea, scala vestibuli, scala tympani, scala media, vestibular membrane, basilar membrane, stria vascularis, helicotrema, modiolus, spiral ganglion, spiral lamina, spiral limbus, organ of Corti, tectorial membrane, hair cells, cochlear nerve and vestibular nerve.